

**Woolworths Online Order and Delivery System –
Enterprise Systems Analysis, Design, and Modelling
Project-II**

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Summary of Systems Analysis

From the analysis of part 1, a proposal has been conducted for Woolworths New Zealand's Online and Delivery System. Based on some assumptions made. The previous report analyzed the identified key operational problems across 3 departments. These departments include store operations, supply chain & Logistics, and customer service.

The main issue found during the system analysis is that there is currently a heavy reliance on manual processes, which reduces real-time data visibility and efficiency in staff communication. This information is collected through multiple sources, which include the original problem statement, interviews with stakeholders, and, lastly, analyzing legacy systems. This directly helped produce the business goals, user requirements, functional requirements, and non-functional requirements.

In part 1, both the initial class diagram and use case diagram are created. The use case diagram includes both the customer-facing use case and another use case diagram for the internal management system. The initial class diagram covers classes related to customer, staff order, inventory, delivery, and reporting. Upon creating this, several weaknesses are found in the class diagram, including the absence of abstract classes and interfaces, and redundancies.

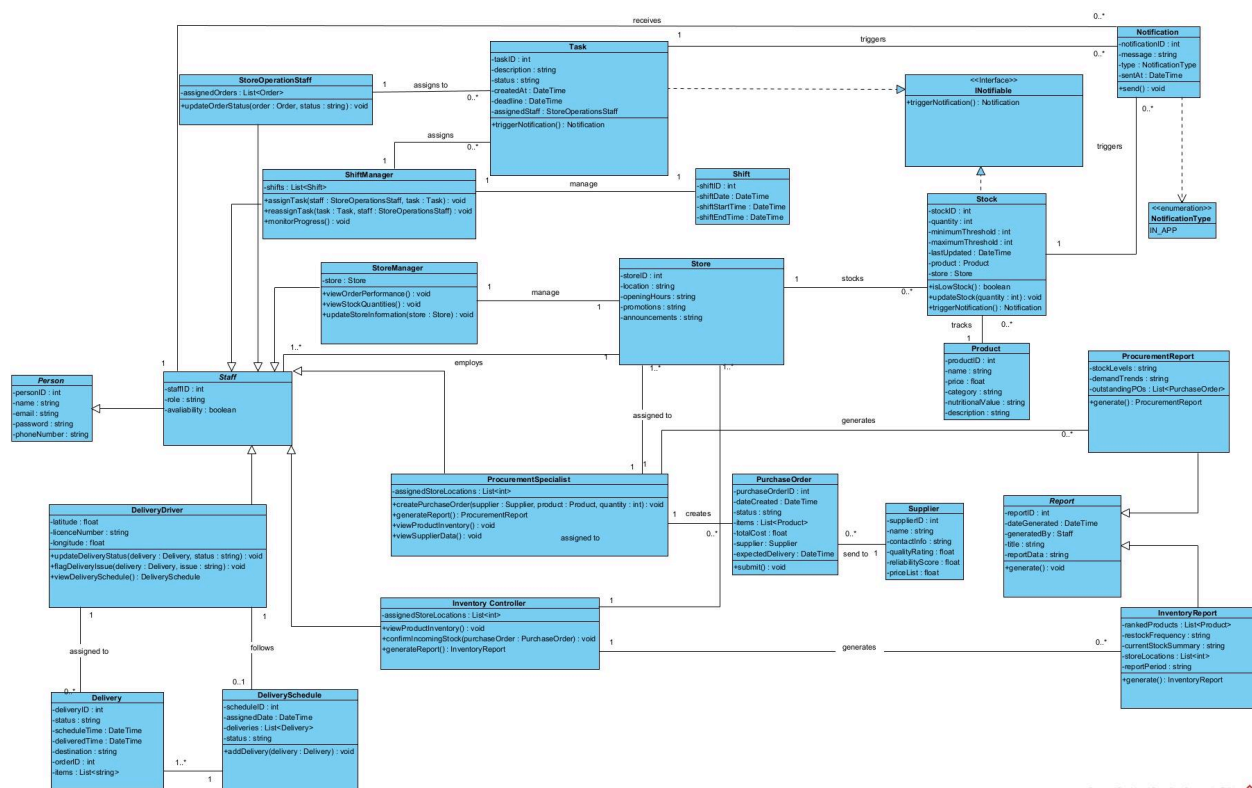
Scope Refinement

Continuing this report, the scope has been reduced to just focusing on the internal management system, and the customer-facing Online platform is out of the scope. Customer and customer service representative functionality has been excluded due to time constraints. Additionally, delivery management has been moved from the supply chain & logistics department towards the customer service department. The scope covered within the internal management system would be the Store operation staff, Shift manager, store manager, procurement specialist, inventory controller, and delivery driver.

Refined Static and Dynamic Modelling

Refined Class Diagram

☒ ~~Refined UML Class Diagram with abstraction and interfaces~~



This updated iteration of the class diagram shows:

- **Abstract classes of Person, Staff, and Report (Italic)**

Since these classes are never initialized directly, Person and Staff are base classes for the subclasses of staff. Report is the base class for the Procurement report and the Inventory report class.

- Enum NotificationType

Changed from a string attribute called "type" to adding an enum "NotificationType", even though there is just 1 constant value in NotificationType, it would be helpful for future different types of notifications, for example, email or SMS.

- **Changed currentLocation**

From the risk identified from phase 1, the `currentLocation` variable in the `DeliveryDriver` class is split into Longitude and latitude to enable exact distance calculations for delivery scheduling.

- **Added Interface**

An interface has been added to enforce methods on multiple classes. The interface called “`INotifiable`” is used for enforcing the `triggerNotification` method on both the `Task` and `Stock` classes.

- **Scope Reduced**

As stated in the Scoper refinement, there will be no customer-facing functionalities, thus customer, cart, shoppingList, `ShoppingListItem`, Customer service representative, complaints, and refund.

Sequence Diagram

- ☐ Minimum of three Sequence Diagrams

Activity Diagrams / State Machine Diagrams

- ☐ Minimum of 2 activity diagrams or state machine diagrams

Design Pattern Application and Implementation

Application

- ☐ Application of at least two GoF design patterns
- ☐ UML diagrams showing pattern structure

Implementation

- ☐ Corresponding C# implementation demonstrating each pattern

Comparative Design Analysis

Students must provide a critical comparison of the system design before and after the application of design patterns, addressing the following aspects:

- ☐ Changes to class structure and responsibility allocation
- ☐ Improvements in cohesion and reductions in coupling
- ☐ Impact on extensibility and maintainability
- ☐ Consequences for C# implementation complexity
- ☐ Examples of how new requirements can be accommodated more easily

The analysis must be supported by UML diagrams and concrete references to the C# codebase.

Critical Reflection

Design Pattern Tradeoffs

- ☐ Design trade-offs introduced by design patterns

Limitations of the final design and implementation

- ☐ Limitations of the final design and implementation

Lessons learned

- ☐ Lessons learned from aligning UML models with C# code.

References

Appendices

- GitHub
(<https://github.com/ShawnLeeyz/Woolworths-Systems-Analysis-Design-and-Modelling-Project-I.git>)